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Sensors:

PRODUCTION CAN NOW GET CONTROLLED By Products Themselves

Wireless sensor networks as an enabling technology for the The Internet of Things (IoT) is used in virtually all industries. Thoralf Dietz, head of corporate communications, Fraunhofer Institute for Integrated Circuits IIS, gives us a glimpse into cutting-edge sensor technologies, in a conversation with Dilin Anand from EFY

Q. Any recent development in sensor systems technology at your institute?

A. There is a new 3D magnetic field sensor technology by the name HallinOne. It allows for three-axis measurement of any magnetic field from one spot in a device such as a washing machine. HallinOneFH5401 has two 3D sensors so that even gradients of the magnetic field can be measured. This reduces the influence of external disturbing magnetic fields in most applications. It is also well-suited for angle based measurements and almost eliminates the effects of changing temperatures of the magnet.

Q. What about developments for critical verticals such as healthcare?

A. We have developed a wireless system for locating medical equipment and blood products as well as for condition monitoring. The project called OPAL-Health makes managing medical equipment easier and improves the safety of transfusion procedures by smartobjects—microelectronic modules that can gather and save information from their environment and communicate wirelessly.

Q. How have your institute's wireless sensor network technologies improved in the last two years?

A. By adaptive frequency hopping because of a fully-decentralised channel-selection scheme, it can deal with disruptions without any expensive

measurement campaigns and frequency assignments in advance.

Further, sensors remain in their energy-efficient synchronous operation even when not within reach of an infrastructure network. This is the key to a robust sensor network covering wide geographic areas or for Industry 4.0.

Q. What are the most recent advances in s-net technology?

A. Application of s-net technology in the production industry combines wireless communication, tracking and sensing with application and process logic on a single platform. This allows for connected processes and precise location of individual objects or components at factory sites. In smart production, production can now get controlled by products themselves. We can get comprehensive sensory monitoring in smart environments, along with a combination of gapless traceability in smart tracking along with sensing and local application logic.

Q. As sensor technologies improve, what new applications are these capable of driving?

A. Products in the home automation sector are already available, and new applications are possible in transport and logistics. Wireless sensors with BLE or even tracking systems (like Fraunhofers Daedalus project) for small goods like boxes are

powered with energy harvesting. In the nearer future, wearables like human body sensors will be powered, or at least assisted with energy harvesting.

Q. What are the latest improvements to signal processing, circuit technology as well as mixed signal IC design used in integrated sensor system technologies?

A. In addition to HallinOne, there are many improvements in increasing sensitivity and offset reduction. But the most important progress is that we can make the final test of magnetic field sensors after fabrication without any external fields and therefore without a magnetic handler. This means precise 3D magnetic field sensors at a very low price.

Q. What are the most exciting projects currently in R&D and expected in the next few years from sensor systems research area?

A. For hall sensors we work on a 3D magnetic field camera, which is fully calibrated. Calibration of the three magnetic field components is state-of-the-art, but calibration of orthogonality of these three components to each other is a challenge.

Moreover, in our Übermorgen-Projekt (projects for markets beyond tomorrow), SENEKA, we work on solutions that allow effective use of technical devices for disaster management. **EFY**